

Mohamed Zaitoun

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EDUCATION

Purdue University

Bachelor of Science in Computer Engineering — Dual Degree Program

West Lafayette, IN, USA

Aug. 2025 – May 2026

Egypt University of Informatics (EUI)

Bachelor of Science in Computer Engineering — Dual Degree Program

Cairo, Egypt

Oct. 2021 – May 2026

EXPERIENCE

Undergraduate Research Assistant

Purdue University, CYNICS Lab

Jan. 2026 – May 2026

West Lafayette, IN, USA

- Conducted research on secure audio transmission, investigating AI-based compression and obfuscation techniques for efficient and privacy-preserving communication, with performance goals comparable to established Opus codec.
- Developed **Enadox**, an end-to-end encrypted messaging prototype with a **C#/.NET** desktop client, Win32 API integration, local language-model inference, and a secure **FastAPI** backend.
- Built a local compression engine within **Enadox** using a quantized **RWKV** language model to compress plaintext payloads before encryption, alongside **Python-based tooling** for experimentation, benchmarking, and system evaluation.

Undergraduate Teaching Assistant

Purdue University

Aug. 2025 – May 2026

West Lafayette, IN, USA

- Served as a primary point of contact for students in **ECE 570, ECE 473, and ECE 302**, holding office hours and troubleshooting complex coursework and programming issues.
- Owned assignment and project evaluation workflows, applying standardized rubrics to provide fair, consistent, and timely feedback while partnering with faculty to improve course materials.

Fullstack Software Engineer

Aallie

Dec. 2024 – Aug. 2025

Riverside, CA, Remote

- Architected and shipped core features for an AI-powered CFO platform, integrating **OpenAI APIs** as the primary LLM for agentic financial workflows, forecasting, and real-time analytics using **Laravel, React, Redux, MongoDB, and PostgreSQL**.
- Led end-to-end development of a pool-construction project management platform, automating scheduling and workflow logic with **Node.js, TypeScript, and Prisma**.
- Designed relational and non-relational data models for data-intensive financial reporting, improving consistency and performance across **PostgreSQL and MongoDB** workloads.

Fullstack Software Engineer Intern

Aallie

Sept. 2024 – Dec. 2024

Riverside, CA, Remote

- Built responsive **React** components and secure RESTful API endpoints for the initial rollout of an AI-powered CFO platform, collaborating with senior engineers across the stack.
- Implemented **Redux** state-management patterns and debugged financial data synchronization issues to improve frontend reliability and maintainability.

Fullstack & IoT Engineer Intern

Egyptian National Academy of IT for People with Disabilities

Aug. 2024 – Sept. 2024

Cairo, Egypt

- Assembled, programmed, and tested an IoT hardware prototype integrating software components with live sensor data, then demonstrated the system to stakeholders.
- Translated hardware constraints and accessibility requirements gathered from end users into technical specifications for assistive technology.

Software Engineer Intern

Nexus Analytica

Aug. 2023 – Oct. 2023

Cairo, Egypt

- Contributed to an internal energy-data analysis tool and designed an Entity-Relationship model to restructure heterogeneous datasets for downstream analysis.

TECHNICAL SKILLS

Languages: C++, C#, Java, Python, TypeScript/JavaScript, PHP, SQL

Frontend & Backend: React, Redux, Zustand, Node.js, NestJS, Next.js, ASP.NET Core, Spring Boot, JPA, Hibernate, Laravel, FastAPI, BullMQ, RESTful APIs, GraphQL

AI/ML: Generative AI, RAG, PyTorch, scikit-learn, Hugging Face, LangChain, LLM APIs (OpenAI, Google Gemini, Anthropic), MCPs

Databases & Cloud: PostgreSQL, MongoDB, Redis, Kafka, AWS Fargate, AWS Lambda, AWS SQS, AWS CloudWatch, Azure Web Services, Vercel, Railway, Docker

DevOps & Tools: Git, GitHub Actions CI/CD, Linux, Bash, PowerShell, Claude Code, Codex

PROJECTS

MingleMart | *Java 21, Spring Boot, Spring Modulith, React 19, PostgreSQL, Redis, Docker* Jun. 2026 – Aug. 2026

- Built a full-stack e-commerce platform with a **Java 21 Spring Modulith** backend and **React 19** storefront, implementing catalog, authentication, user profiles, offers, reviews, and notification workflows.
- Designed a modular backend with isolated bounded contexts and contract-based inter-module communication, enforcing architectural boundaries through automated **Spring Modulith** tests.
- Implemented secure authentication with **Spring Security**, **JWT access tokens**, **rotating refresh tokens**, **Google OAuth**, **BCrypt**, and single-use email verification/password-reset tokens.
- Built a hybrid **SSR/CSR** frontend with route-level data prefetching and an anonymous **LRU page cache**, while using **Redis** for resilient storefront caching and PostgreSQL/Flyway for transactional persistence.

SURGE | *FastAPI, React, SQLAlchemy, PostgreSQL, Celery, Redis, LangChain, Gemini* Jun. 2026 – August. 2026

- Built a multi-tenant B2B outreach automation platform for importing leads, segmenting audiences, and executing AI-generated email campaigns end-to-end.
- Designed asynchronous campaign execution with **Celery** and **Redis**, integrating **LangChain**, **Google Gemini**, **Apollo.io**, and **Resend APIs** for personalized generation, lead sourcing, and delivery.
- Developed the backend with **FastAPI** and **SQLAlchemy/PostgreSQL** and the frontend with **React 19**, enforcing tenant-level data isolation across business clients.

CodeAtlas | *TypeScript, NestJS, React, TypeORM, PostgreSQL, LangChain* May 2026 – August. 2026

- Built an AI-powered developer tool that analyzes GitHub repositories and resumes to generate tailored project documentation and READMEs.
- Orchestrated agentic workflows with **LangChain** and automated repository updates by opening pull requests or committing generated READMEs directly to GitHub.
- Implemented the backend with **NestJS** and **TypeORM/PostgreSQL** and the frontend with **React 19** and **TanStack Start**.

CiteMind | *Python, PyTorch, Hugging Face, Reinforcement Learning, Whisper, BART* Feb. 2026 – Apr. 2026

- Engineered a multimodal audio-language model for academic citation recommendation by connecting a frozen **OpenAI Whisper** audio encoder to a **BART** sequence-to-sequence decoder through a custom cross-modal layer.
- Designed a two-stage training pipeline on **A100 GPUs**, combining masked generative pre-training with PPO-based reinforcement learning and a composite reward based on NLI entailment and semantic similarity.
- Curated an **80,000-sample** dataset through the Semantic Scholar API; RL fine-tuning reduced citation hallucination from **11% to 2%** and improved format accuracy from **89% to 98%**.